Midterm 1 - Fall 2026

ECON 4753 — University of Arkansas

1. This question concerns our general discussion of modelling outcomes. Say we have an outcome variable Y that we want to analyze and a set of variables X we want to use in our model. In general, we write the 'true model' as $Y_i = f(X_i) + e_i$. With a sample, we will fit some model which we will call $\hat{f}$.

   Two models were fit using the same training sample. Model A is a simple linear regression, while Model B allows a more complicated relationship between X and Y.

Model	In-sample MSPE
A	420
B	310

   i. [10 points] Write the formula for in-sample mean-squared prediction error.

   Answer: $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$

   ii. [10 points] A student argues, "Model B has a lower in-sample MSPE, so it will predict outcomes more accurately for new observations." Do you agree with this conclusion? Explain.

   Answer: I disagree with this conclusion. Using the in-sample MSPE to choose a model is problematic becuase of the potential for overfitting. While a lower MSPE can signal a model that better predicts $f(X_i)$, it could also signal that a model is overfit. In general it is better to use the out-of-sample mean squared prediction error.

2. This question is based on our review of statistics. A company advertises that its wireless earbuds last an average of 7 hours on a single charge. You independently test a random sample of $n = 64$ pairs of this model under the same conditions. Let b_i denote battery life in hours. Your sample has a mean of $\bar{b} = 6.6$ hours and a variance of $\text{var}\,(b) = 4$ hours squared. The sample distribution of the sample mean is therefore $\bar{b} \sim \mathcal{N}(\mu, 1/16)$, where μ is the true average battery life.

(a) [10 points] Form a 95% confidence interval for the population mean battery life. Interpret this interval in words.

Answer:

$$\left(6.6 - 1.96 * \sqrt{\frac{1}{16}}, 6.6 + 1.96 * \sqrt{\frac{1}{16}}\right) = (6.11, 7.09)$$

With 95% confidence, the true average battery life falls in the range 6.11 to 7.09 hours.

(b) [10 points] You suspect that the average battery life is less than the advertised 7 hours. Test the null hypothesis that $H_0 : \mu = 7$ at the 5% significance level.

Answer: Since the value 7 falls inside my 95% confidence interval for the sample mean, I fail to reject the null at the 5% significance level.

3. The following questions will focus on a dataset from Spotify. A random sample of songs were drawn from 6 genres: 'EDM', 'latin', 'r&b', 'pop', 'rap', and 'rock'. Spotify produced a few variables we will use for the analysis: (1) `track_popularity` measures track popularity on a scale from 0 to 100 and (2) `danceability` measures how suitable a track is for dancing on a scale from 0 to 1.

   I ran a regression of track popularity on the song's danceability measure.

```
OLS estimation, Dep. Var.: track_popularity
Observations: 32,833
             Estimate Std. Error t value  Pr(>|t|)
(Intercept)   35.1757   0.632777 55.5894 < 2.2e-16 ***
danceability  11.1497   0.945354 11.7942 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

   i. [10 points] Interpret the coefficient on a song's `danceability` measure

   Answer: For every 0.1 units a song is higher on the `danceability` measure, the track's popularity will improve by $11.1497 * 0.1 = 1.15$ points.

   ii. [10 points] Form a 95% confidence interval for the coefficient.

   Answer: With 95% confidence, the slope coefficient falls in this range:

$$(11.1497 - 1.96 * 0.945354, 11.1497 + 1.96 * 0.945354) = (9.30, 13)$$

4. Continuing with the Spotify dataset, we want to see if different genre's have different popularities. We regress the track's popularity on indicators for each genre (excluding one)

```
OLS estimation, Dep. Var.: track_popularity
Observations: 32,833
               Estimate Std. Error   t value  Pr(>|t|)
(Intercept)   47.744870   0.339019 140.83227 < 2.2e-16 ***
genre::edm   -12.911344   0.451279 -28.61054 < 2.2e-16 ***
genre::latin  -0.718294   0.490232  -1.46521   0.14287
genre::r&b    -6.521339   0.488259 -13.35632 < 2.2e-16 ***
genre::rap    -4.529416   0.457639  -9.89736 < 2.2e-16 ***
genre::rock   -6.016532   0.489295 -12.29632 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

i. [10 points] What is the omitted group in this regression?

Answer: The omitted group is pop.

ii. [10 points] Interpret the coefficient on rock in words.

Answer: On average, rock songs have an average track popularity 6 points lower compared to pop songs.

iii. [10 points] Which genre is more popular, r&b or rap?

Answer: R&b songs have average popularity $47.74 + (-6.52) = 41.22$.

Rap songs have average popularity $47.74 + (-4.53) = 43.21$.

Therefore, rap songs are more popular on average.

iv. [10 points] How would you modify this regression to test whether this difference is statistically significant?

Answer: To test whether this difference is statistically significant, I would rerun this regression and make rap the omitted group (or r&b!).